

Uncertainty and active inference - Thomas Parr

keynote at HLAI

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thank you very much for the introduction and say yes I'm going to talk about active inference which is a theoretical framework we use to try and understand brain function and the neurobiological processes that underwrite a lot of the cognitive functions that were interested in and the the kind of central idea and active inferences to treat everything the brain does as an inference problem of some sort and that includes that includes perceptual inference most obviously but also for the notion of planning as inference and and movement itself as an inferential process fulfilling addictions about the world I'm going to talk about a particularly the role of uncertainty in these kinds of formulations because this is one of the things that active inference does quite well and those of you who are in the last session of or lunch will probably recognize a lot of the concepts that come up here so the basic structure is I'm going to first of all give an overview of the idea of active inference and just a brief summary of what it is I'm then going to talk a bit about the sort of neuronal networks the kinds of architectures we might expect biological systems to use to actually solve these inference problems and then I'm going to give you a couple of examples and show some simulations that illustrate the behavior of this kind of system first one looking at the role of uncertainty the idea of exploring to maximize my information gain in the environment and the second two instead of thinking about uncertainty about what the states and environment causing my sensations actually thinking about what what influence does mine certainty about the probabilistic structure of the world and what influence does that have over the behavior as I choose to engage in so they're thinking about the role of learning as opposed to inference yes okay okay and right can you hear me now it's that okay yeah okay I'll try and let me know if I drop too quietly again and the final thing I want to talk about will be the processes that convert and the decisions we make and the plans that though we infer we should be following in two movements in the real world and I'll give you some examples for each of those saying the key notion or the key

objective function of active inference the key thing that everything works to try and optimize is this free energy functional here and this is something that comes up in variational inference this is just a quantity that says that if I have some beliefs this as its distribution Q about the states s that are causing things in the world and then I can I can plug them in to this function and by minimizing now function and I I can then get the best set of beliefs that best describe what's going on in the outside world that depends upon there being some sort of probabilistic model that I have about about the way the world is structured and that's given in this distribution P so that's that's saying how how the states s give rise to observations oh and little the till to the line over the top indicates that these are trajectories we're interested in a process that's evolving over time they'll determine here the pie that everything is conditioned on is the the particular model of how things are going to evolve all the policy that I'm going to assume so this essentially says a policy is just a sequence of actions or a sequence of states that I can I can follow and I have several different models for how I could behave several different alternative policies I could engage in and I can associate a free energy functional with each of those that is fine when I'm trying to make inferences about the data I currently have the observations that are currently available to me it becomes much more problematic if I then want to move into the realm of planning because if I want to minimize my free energy in the future I need to try and predict what that free energy might be and you can see here the free energy as it is is a function of observations of data as I don't have data for the future yet by definition I instead replace the expectation here or the distribution I'm using to average this with one that includes the observations so I'm now predicting what sort of observations I'm going to get and I'm averaging this thing with respect to both the observations and the states that I expected to delete them and that gives us something to think about in terms of how we plan and how we are we perceive these things into into the future so once we do this we can think of perception is the process of optimizing this distribution q what are my beliefs about the states of the world at present and what will they be in the future and and I can optimize that and I can simply plug that in to the expected free energy the tobo I've defined here I'll come back to that in a second and I can use that to to form beliefs about the sorts of plans I'm going to engage in the sort of policies I'll pursue so here this is just the policy of the belief I have about how I'm going to act I can use that to generate an action I'll come back to that as well and that action changes the world generates new observations and we get this kind of action perception cycle with planning and opposed then now the reason I've put these two equations on is just to emphasize some important features of these two Ferengi functionals and

what they may actually mean and here you can see that we can react spresso thing here in terms of this thing here which is just the negative log probability of a certain set of observations and you can think if this is the evidence for the model of all the evidence that I'm continuing to exist I'm seeing sort of data that are consistent with with my own model of myself the bill is actually changing with respect to this Q distribution is this bit here this KL divergence this is a non-negative quantity since it's always positive that just quantifies how different these two distributions are and here we're saying that this is the posterior distribution the probability of States given the data I've already observed and and this is our beliefs so essentially by minimizing the free-energy and I'm effectively reducing the distant or I'm trying to approximate what the true posterior should be as best as possible using my current beliefs of course if you were to just substitute in the true posterior if you could calculate that analytically then this would just become exact inference I point out that it depending on what field you come from sometimes the pre energy is represented as the negative of what it is here particularly in machine learning and that often leads the rhetoric of maximizing free energy or maximizing the evidence lower bound I'm going to be talking about it using the kind of physicists definition which involves minimizing the free energy so this is now an author pound on the negative evidence this here is the expected for energy rearranged in a similar way and you see we still get this kind of expected log evidence term and we can think of this as being a kind of goal directed drive this is saying that I want to seek out though those data that are consistent with my model and those might be those data those might be introspective data that consistent with maintenance of homeostasis or something like that they could also be extra step to the no goal-directed task of some sort the interesting thing here is that the expected KL divergence here is swapped round just so the sign is now negative and we now have the expected posterior and the current posterior and the reason for that is the way we take the expectation here and that means that the to minimize the expected for energy I need to actually maximize the difference between my beliefs as they currently stands this Q distribution here and what I expect my beliefs to be after making the observations that give rise to this posterior here so now we have an interesting and interesting way of thinking about what this is there's a drive here to fulfill the kind of goal directed predictions and there's a drive here to try and change my beliefs as much as possible by observing new data so we can think of this as carving up the expected free energy into a component that drives exploration information seeking behavior very much as many of the terms did in the in the previous talk and a drive towards exploration and we can think we can give these labels like intrinsic and

extrinsic value and embarrassing by national I'll come in a moment to the relationship between this and various other different kinds of scheme first of all just to make it explicit what this looks like when when when you actually minimize the free energy when you work out what the posterior believes should be so this is assuming a sort of a partially observed Markov decision process like a form where policies cause state transitions and states cause observations and so when we minimize the free energy we essentially get my belief about states the present is a function of beliefs about States the past and some transitions beliefs about the future and some transitions and the evidence I'm getting at the moment so essentially we combine two empirical priors with with sensory evidence or evidence for data and combine those to try and form a posterior belief about the present and that's our that's our perceptual inference that minimization of the the free energy when we when we look at the realm of planning if we have a private leaf that we minimize the expected for energy the posterior belief will contain both the expected for energy and the free energy so here we have another yet another expression of the free energy and the expected for energy and I'm just going to write out each term in that explicitly and just talk you through how you can intuitively think about what this means and what is doing the first time here we generally refer to as accuracy so this is essentially saying given I have several beliefs about what sort of policies I could engage in and I can I can take the data I've already gathered and work out whether that carries evidence in favor of me pursuing one policy or pursuing another and so here it's just saying the observations I've observed along the course of following this sequence of actions is consistent with me having followed that sequence of actions then it's probable that I am following that sequence of actions this term here oh well let's go in sequence so so this term here is the sort of thing that's used in approaches like uncertainty sampling looking for those data about which I'm most uncertain so H here just means the entropy so if I expect something to be very very uncertain in a particular place then it means I'm more likely to try and try and observe those data and learn more about the world if we combine these two terms so here we now have a conditional entropy that says how reliable is the mapping from States to observations and how informative will they be if we combine these two terms then we get a and something that's been given a whole range of different names it's sometimes been referred to as information gain you find this in even in the fifties literature on base in experimental design as the value of an experiment based in surprise salience novelty epistemic value and and sometimes an intrinsic value and so this is our information seeking component the final bit here is just to point out that these two things together limits how far my beliefs about about the policy I can

pursue deviate from my prior beliefs if they deviate a long way then as penalized it's an excessively complex explanation for how I could pursue this behavior and final thing is if this term at the end the kind of expected evidence term and we can interpret this in terms of an expected utility this is the thing I'm trying to seek out or in terms of an extrinsic value okay so next thing to talk about is just to think about the source of neuronal architectures that can actually do this sort of thing and the way we get to that is we think we first will a particular model in this case I'm sticking with the Markov decision process model and we express the free energy in terms of the sufficient statistics of those distributions and the parameters of those so here we're dealing with categorical inferences so we're just dealing with vectors and matrices of different kinds of quantities so I can be in one state and in the next day then the next stage and there's a transition matrix that determines how I how I go from one to the other if we just take the free energy gradients of that system with respect to the beliefs about States then we get a series of relatively simple update equations we can do the same thing for planning and again we can just express these in relatively simple linear algebra and again these are just the free energy gradients with respect to each of these things and we just expressed it as a gradient descent on the variational free energy or in this case we set up a belief that I'm going to minim or I'm going to assume those policies that have the minimal expected for energy the nice thing about these equations is that the dependence is between the variables actually are quite similar to the neuronal networks that we that we know are present in the brain or things of a similar form so just to give you a sense of that we know that we we have sensory data that's that inputs to normally granular layer four of the cortex I know that's three in this particular example but we can think of those as computing a prediction error that's the difference between the Leafs I currently have and the sensor in which I have and I can and and you can see the that prediction error also depends upon and this future and past components here and representing past on the Left present future so influence from the past inference from the future influence from sensory data and those errors then being used to drive that gradient descent process I can then evaluate the expected for energy based from that workout with the expected for energy is and then compute the probable policies and here's a basic model average to work out what are the probable states that I could be experiencing and pursuing based upon and based upon my beliefs about the states under each policy and my beliefs about the probability of perceiving each policy okay so that's that's the sort of architecture we would think of and this is similar to the sort of culture code subcortical loops that we expect to find in the brain and this gives us a way to start to think about how we can

interpret mirror Anatomy in terms the inferences that it's trying to pursue with inferences about what's actually out there and inference is about how how am I going to act and what will that do to change what's out there right so next thing I want to do is to give you an explicit example of this in a deliberately very simple generative model and try and give you some intuition for how that works so again this is the partially observed Markov decision process here we have our beliefs about the policy which influenced the transitions from state to state we have transition matrices here let's say well if I'm in a current state what's the probability of being in that state and similarly we have a mapping here that goes from States to observable outcomes and says what's the probability if I'm currently in a particular stage of observing a particular outcome and the key thing here is we can actually parameterize these such that we can alter the uncertainty in each of these distributions and see what effect that has the for exploratory behavior so here in this case imagine I'm in a particular state and each of these represents the probability that if I'm in that particular stage I'm going to observe a particular outcome so here are all the different outcomes I could observe here we've just set up a parameter that we've to notice our Z term or Zeta the effectively inverse temperature parameter on this distribution so we can make it more or less precise if it's very large then this becomes almost deterministic if it's very small then this becomes completely flat I could observe anything based upon where I am we've also done the same thing for transitions here with such parameter that essentially quantifies how uncertain we are about given where I am where I will be next again if it's zero this is very flat and then as we increase it it gets very large and that just means that we can control the uncertainty here what sort of volatility of the sequence and the uncertainty here in the mapping the simulated agent believes occurs between States the world and the observations this is a you know fairly arbitrary way of doing this we could obviously parametrize this in lots of different ways it's just to give you a sense of one way in which we can do this so this is a representation of the generative model that I'm actually going to show the simulation for so here I'll go back a second so here if you imagine what each of these each of these states is a place I can be looking so I can move my eyes to various different locations and fixate different locations in visual space and at each of those I can see something I can get some sort of visual or proprioceptive feedback from where I've ended up looking and so this just becomes the policy just becomes the sequence of saccades the sequence of fast eye movements I'm going to assume in order to induce these state transitions so here we have the hidden states that are the fixation location so these are all guesses so I can I can be fixating in location one two three four or in the center I have another

set of hidden states that I have no control over which are which of these three squares is present in each location so it can be either 1 2 or 3 if I look to location 1 and the third the third of these squares is there then that will map to a visual outcome that says I'm looking at this particular color there are transitions between these and we can control how deterministic or how stochastic those transitions are and we can control the mapping between these things so here these are the those parameters saying how uncertain is the mapping from stage to observation finally we have this this other outcome modality this proprioceptive modality that says where am I currently looking and so these can transition I have control over how these transition together these make up the visual and very perceptive outcomes and the uncertainty of these transitions and the uncertainty of this mapping can all be manipulated and the reason for doing that is just to show you what the behavior of the system is when will it be like so here we have three examples of that same environment but with slightly different slightly different uncertainties in play so here we have the simulated eye tracking trace is interpolated between the points is just choosing one point or the next so it's choosing to look there there there and exploring the environment I should point out there's absolutely no preferences in play here there's no rewarding location everything here is intrinsically motivated through the this drive information game so you can see in this example in every single location the the mapping of the likelihood that says given a stage what's the probable observation is the same and the same is true for the mapping from state to state so everything is equivalently precise or equivalently valid on here just to illustrate what happens we've decreased the precision of the inverse temperature the maps States to observations so now we have a setting we have a square here where if I look in this location the mapping from States to observations is almost completely random or with it's very small relative to relative to the others but that essentially means that the amount of information I can get by looking there about the stage that's actually there is relatively little so here in this presence of you know relatively noisy mapping there's not really much point looking there because the amount of information I can gain in our state's the world is pretty limited and you can actually think of this as being a bit like a darkroom if I look around the environment I'm not actually going to be able to have any sort of precise mapping in play and you can see that it just completely ignores that location and seeks out those where there is a lot of information to be had and some in psychology refer to this is the streetlight effect which is based upon the idea that if you're out late one night and you've had a few drinks and you've dropped your car keys on the street the first place you look for them is under the streetlight not because it's the most likely

place for them to be but just because it's the only place you can actually get any sort of precise information which is intense based optical maybe a few simulators in this sort of setting this final example we've decreased the precision or the inverse temperature of the transition happening so here things transition much more randomly and you can see that in the actual environment so the agent believes it but it's also true of the environment itself and you can see that this location is actually fixated much more often contrary to what's going on here so in this particular scenario you can think of this heuristically as I haven't looked there in a while it's more likely to have changed to many of the others so there is more uncertainty to resolve there if I haven't looked there recently and compared to the others which will remain relatively static or Alize relatively predictable even in the absence of me looking at it this particular location is likely to change and in an unpredictable way so I better look back there and check okay and the reason I put this up here is you can interpret these parameters in terms of which components of this drive to explore have been altered so here we've got the conditional engine yes it's changed by changing this and you can see the dots that essentially gets larger that becomes a less probable policy and similarly for the the change here in the opposite direction okay so the focus there in that particular example was on the uncertainty I might have about states of the world the next thing I'm going to talk about is the uncertainty my have in the agile mappings between things in the the parameters that govern that rather than the states themselves so again we have the same we have the same network here but now we're going to parameterize these mappings the a mappings that go from States to outcomes and we're going to say that this is something that can be learned through the course of the trial or across several trials and it's this parametrized essentially using a drill a distribution and here is an example of the kinds of updating that you might expect from the drill a distribution in the scenario so again this is if I'm in a particular stage these are all the different observations I could make while there if I make this sequence of observations here but every time I see one I update that element essentially of that probability distribution so more likely to see this in future the same thing is true here and the difference between each of these is the choice of the initial drill a parameters so I can actually say how certain am I in this mapping a priori both of them start with the belief that the mapping is itself uniform but I can be incredibly certain that this uniform or I can be relatively unconfident at its uniform and as an analogy if you imagine you flip a coin a hundred times and it comes up heads half the time then you're relatively confident it's a fair coin and a new observation is not going to change your beliefs a great deal whereas if I've only flipped at once or twice before I've go ahead once and tails once I'm going to start

with the same initial expectation of every being fair based upon those data but a new observation will actually cause a much bigger change in my beliefs and that becomes very important in the sense of information gained about parameters because if I'm going to behave in such a way that I minimize my expected free energy or maximize the difference between my current beliefs power parameters and my beliefs in the future then it makes sense that I should I should look to those regions to have a profile more like this compared to those like this because here I'm going to change my beliefs a great deal more than if I make an observation in a region like this and I'll come back to that in the example so the example I have here is we have a state base that is just a grid world so I can look at any location in this in this world it's just me I ain't grid and the state is where am I currently looking in it I can then transition between those and at each stage I can observe one of three things I can observe nothing I can observe a black dot or I can observe a red dot and the way that this is set up is that every time I look at a particular at a black dot it will change color so will change from black to red this isn't explicitly in the generative model this is just a feature of the environment and I can then simulate the same sort of thing by using those same gradient descents on the on the variational free-energy and here you can see it's looking around the world is it's finding these these dots and then it's moving on to the next one it's not looking back at the previous one it's continuing to explore now the motivation for this particular task is partly due to my interest in thinking about how generative models can be broken in the sort of ways we can explain pathology in terms of this but they also give you a nice way of thinking about what sort of was what sort of influences go on in this generative model and and in particular there's a clinical condition called visual neglect where patients with a stroke to the right side of their brain normally and will often neglect the left side of space they'll perform viewers are cats to it so these are attracting traces just at rest looking at neglect patients and and a control group you can see that they perform many many more saccadic eye movements to the right side of space than the same patients will will fail too if you ask them two o'clock they'll draw the right-hand side of it and not the left and there are a range of different phenomena that the Toccoa it is an interesting condition that I went going to too much detail about it now the crucial thing is that that a common test that's used to assess neglect is to give people a sheet of paper with a series of lines drawn rich and to ask them just to cross out all of the lines and they'll often cross out all of those on the right rather than the left the reason I'm talking about this is that and it's an interesting problem thinking about the influence is over exploratory behavior to think what would lead a visual search to towards counseling only those lines on the right faces the left or

Oh or any attending to those on the right versus the left what would lead to this sort of scan pattern in terms of what would have to be broken in the generative model and so here are some examples so here we have the same thing I was showing you before and it looks at both sides roughly the same number of times and there's relatively even but then I have three ways in which we can break this by changing a prior belief in the generative model to behave in an abnormal way the first of these is very simple we just say sorry this this shouldn't be an expectation this is function E and if we if we set up privately if the time more likely to perform saccades to the right from the left it's an almost trivial example but it shows one way in which we can we can produce this sort of a vis just a priori more likely that all I'll hold up over there the second thing we can do is change the model such that there's a bias and proprioceptive leads are more likely to experience data that says I've looked to the left oh sorry to the right than to the left and by minimizing my free energy either I fulfill those predictions and I end up looking to the right more often than to the left but here is probably the most interesting one this is the one I was talking about before so we've got our information game term where looking at how much do my expected beliefs diverge from my current beliefs it's added Virgil lost and that's a good good thing to do so in this particular example here all we've done is we've increased the confidence in the initial distribution exactly as I showed you on the previous couple of slides ago we've set it so that your new observation will cause a relatively small change in my beliefs about this mapping where I looked to what I'm going to see so the mapping from what to wear I'm sorry even from where to watch the where I'm looking to which of these targets I see becomes more confident and as if they is modified less after a saccade to the left versus right so if you think of the kind of metaphor of the brain is a scientist who is trying to work out what's the best experiment before next you don't do an experiment that just changes nothing about your beliefs you want to do the sorts of experiments that will get your new data that will will change your beliefs about the world and that's exactly what we're saying here and you can see the influence that's having over behavior again this is just a gradient descent on the free energy gradients but it's giving us this interesting pattern similar to these other types of illusions and it's just a speculative link between some of these in different regions of the brain so we can think about things like this we're altering a mapping as altering synaptic connections in the brain and that fits quite nicely with ideas of long-term plasticity synaptic modulation as a mechanism of further learning what's there right and so I'm going to come to the final bit now which is thinking about given these kinds of structures and give it given these kinds of architectures what we've really

thought about are how do I decide where to look or how do I make decisions of some other type but they're all decisions they're all categorical I can over there or there or there ultimately when you actually engage with the world you have to do it in a continuous way you know we have to engage with approximately Newtonian laws and generate continuous changes in continuous variables and so the final that I want to talk about is is how we actually go about doing that how we can incorporate that into a generative model and again appeal to exactly the same notion of free energy minimization that we have to to frame perception and decision-making so this looks very similar to what I've shown you before but it's actually subtly different and here I've changed the variables so these are now continuous variables where we have some causes in the world some hidden states which in our X and some sensory data but now this is not a progression over time so the thing that's changed here is that now we're looking at the mapping from my position to my velocity my velocity to my acceleration and successive temporal derivatives so I can now express a kind of stochastic trajectory in terms of a sort of Taylor series approximation where I've just brought out all the coefficients of that Taylor series approximation which gives me a way of instantaneously representing a trajectory rather than just a current stage for those familiar with it if you were to truncate this here this would essentially be an extended kalman filter model so if we find the sorry that's all a bit too small but and if we find the free energy gradients for that for that system we can reconstruct a similar network if it relies upon errors here the drive changes in expectations and and then those influence the higher-level causes and what's going on my my errors about the states themselves so this can be extended hierarchically essentially indefinitely so some of you may recognize this as being a predictive coding architecture well where we are just passing prediction errors and expectations around again this is the position velocity acceleration the new aspect of this that wasn't in what I showed you previously is this fit here you probably can't read that very clearly but that essentially says the action now is also going to follow a gradient descent on the free energy the only thing action can change from the perspective the generation model here is the it's the data itself so I can either optimize my beliefs about the data or I can change the data source it becomes more consistent with my model and when we frame it like that we see that with the free energy gradients take the form of essential weighted prediction error and so I can treat action as being I get some sensory data i compute an error that drives an action which then in this particular case moves my eyes such that it fulfills any prediction error that's in play and so we get watch neuroscientists have studied for centuries now the idea of a classical reflex arc okay yeah so what if described here gives you a way of if

you have some private leaves about what sort of trajectories you're going to engage in then you can you can then implement those trajectories but the big question is where where do those prior beliefs actually come from OS shape so here's an example of some a simulation using these equations that the treats the eyes both as spheres the agent can and can change the acceleration or the circular acceleration of the eyes and they obey essentially Newton's second law for rotational systems by adding in a prior belief about the sorts of chariot engages in you can see that the eyes then follow that trajectory and giving a different prior beliefs gives you all sorts of different all different sorts of ocular motor behaviors in this particular example we've used the same sort of thing to simulate things like handwriting and various other types of behavior with exactly the same set of equations but different generative models in the past and yes so the next question was was where do these primaries come from so I have this network if I'm if I if I set some prior belief about V then this will cause a particular change in trajectory which will cause particular data that I will predictions about data to tell then fulfill by acting and but I can push the network that I described above on top of this one so I can I can just say that instead of this being an observation in and of itself this is actually just another random variable the index is which of these continuous variables to make use of so you can think of this as a process of forming different hypotheses that then have to be implemented in the continuous domain different alternative alternative trajectories that I could choose from and I infer which one is most probable based upon beliefs here which includes all of the exploratory and and and pragmatic behavior but then I can map that down into this continuous model with several different hypotheses and then pass messages back up that are effectively the evidence for each of those models so I have a hypothesis that I might look over there this is a hypothesis son I look over there each of those can be mapped to a prior belief that my eyes will end up in this continuous location or this continuous location and and then I can I can allow those continuous dynamics play out evaluate the evidence for each of those and pass that back up to say what was the more probable location that I my eyes went to so he get a process of decisions going down to movements and messages being passed back up again and you start to get this hierarchical structure where you can pass messages both up and down again you can extend each of these indefinitely hierarchically so you can repeat this above itself and above itself and similarly you can have many many layers of this and get fairly complicated and models overall so this is just a summary of what I've said over the previous few slides this idea that we can have these discrete equations that are playing out that has said these discrete inferences that are playing out that is saying what are the most probable policies

I'm going to pursue what are the most probable States the result from that and what are the most probable outcomes and observations that I'll get based upon that these feed down through essentially model averaging a bit like a mixture of gaussians from a discrete variable a categorical variable to a continuous variable we then perform these filtering steps that essentially behave a bit like an extended Kalman filter but with action added on and then we pass that all up again evaluate the evidence for each different alternative model which we can do fairly efficiently using Bayes and mul reduction and pass that up to inform the next step so we can get this kind of complete perception action cycle so we can see how that might be implemented in the anatomy of the brain and we've argued some of these structures more than others that pretty good evidence that things like the basal ganglia might evaluate the evidence of different kinds of policies behavioral Tetris I could engage in on the left we're just showing these some of the inferences that happen a teacher status so this is which the the states under each different kind of policy which location am I going to look at and I'm inferring those both of the present time and a future time steps I then map that through with this bit so this is representing the kind of topography of the superior colliculus in the midbrain and the idea here is that each location in the colliculus represents a different hypothesis about where I could look and this red dot indicates the error between where I'm currently looking and where I think I should be looking and you can see that as we form an inference about the next stage its moves out and as the error increases I now believe I should be looking somewhere else as the eyes move there that error closes in there is a cell population in the superior colliculus the behaves almost exactly like this and so it seems again very biologically plausible okay and so that is the end of what I want to say I just wanted to thank these various people and my funders in university and thank you for listening so anyone have any questions [Music] [Applause]